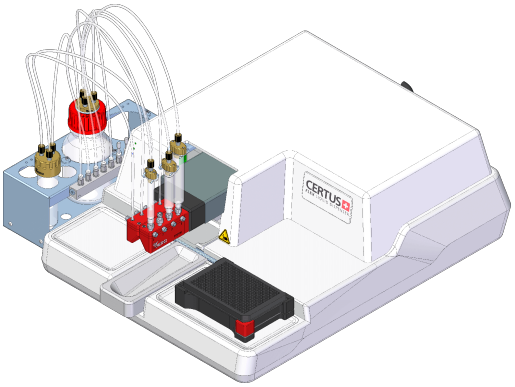
CERTUS
FLEX LIQUID DISPENSER

User Manual

CERTUS FLEX

CERTUS FLEX



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Fritz Gyger AG

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1. Introduction

You have decided for a reliable and high-quality dispenser with highly precise Gyger micro valve technology.

CERTUS FLEX is a fully automatic dispenser for contactless dispensing of various fluids into well plates. Our electromagnetically controlled micro valves enable you to precisely dispense up to eight fluids from 50nl. Automatic dispensing is controlled by **CERTUS CONTROL** software. The unit has been developed to support you with serial dispensing of very small amounts under laboratory conditions.

This knowledgebase will help you optimize the use of **CERTUS FLEX** and avoid mistakes. Most questions are answered in this user manual.

If you have any question or need support, please contact your distributor.

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3. Products

3.1 Certus Flex

3.1.1 About this Documentation

This section provides general information on the use of this user manual.

It also contains important information on the documentation itself, such as the presentation of information, the safety standards and directives applied, and the copyright.

Additional Documents needed

In addition, the following documents are needed to ensure the safe use of the unit:

- Micro valve datasheet, document identification: *CERTUSAN002SMLD300GCDEVxxx.pdf*
- CERTUS FLEX product catalog: *CERTUS FLEX ProductCatalog EN Vxxx.pdf*

General

LANGUAGE VERSION

The original user manual of the unit is written in German. All other language versions are translations of the original user manual!

In case of doubt, the information contained in the original user manual should be prevailing.

ELECTRONIC DOCUMENTATION

The original user manual is available in printout form and electronically in PDF format.

Copyright

All rights reserved. This user manual must not be reproduced or converted, either entirely or in part, into electronic or machine-readable form without written approval from FritzGyger AG.

Responsibility for the documentation

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Maintenance and storage of documentation

To make sure that the user manual is at any time complete and up to date, please do not remove any of its documents. Any new additional documents should be immediately filed. Keep the user manual in an easily accessible place near the unit.

MAINTENANCE

The operational availability and value retention of the unit largely depend on diligent and competentmaintenance.

In addition to its maintenance and care,the unit has to be properly operated if operating personnel and other persons are to remain safe.

QUALITY

We strive to ensure the highest quality and safety of our products. To this end, we request your support:

- Please report to us any residual hazards related to our product which are not described in this documentation.
- Please bring to our attention any error/inaccuracy that you may stumble upon in this documentation.

Presentation of Information

The safety instructions, symbols, definitions, and abbreviations used throughout this user manual are explained below.

SAFETY INSTRUCTIONS

Through safety instructions,this user manual informs about the type of and effects of failure to take measures to avoid hazards.

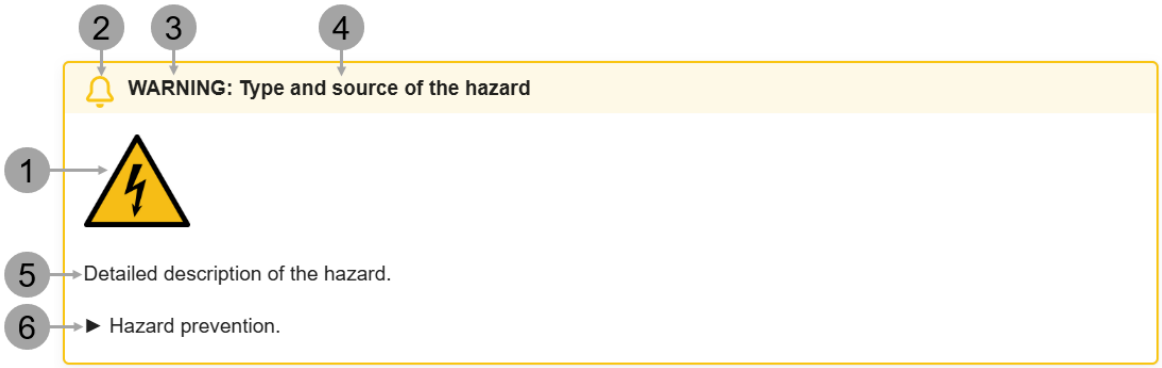

Figure 1 - Safety Instructions Template

1	Warning symbol	Contains a symbol indicating the type of hazard
2	Warning icon	The icon respective to the type of the note, like warning, caution, hazard, danger, etc.
3	Signal word field	Draws attention to a hazard
4	Type and source of hazard	Specifies the type and source of a hazard
5	Hazard description	Detailed description of a hazard
6	Hazard prevention	Measures to prevent a hazard

Document Layout

The following conventions are used throughout the text:

- References to other sections or external documents are highlighted in blue and are often started with the word “See”.
- The dot is used as decimal separator.

Signal Words

The signal word is part of the safety instructions, and denotes the intensity of a hazard.

Hazard

Denotes an immediate hazard. If you do not manage to avoid it, you can die or sustain the most severe of irreversible injury.

Warning

Denotes a possibly dangerous situation. If you do not manage to avoid it, you can die or sustain the most severe of irreversible injury.

Caution

Denotes a possibly dangerous situation. If you do not manage to avoid it, you can sustain a lilght or minor injury.

Note

Denotes a situation that can result in damage, production loss, or an environmental hazard.

Important

Indicates a situation requiring attention to ensure proper operation and maintenance of the unit.

Tip

Indicates a useful hint to make the work easier or more efficient.

Standards and Directives applied

The unit is the state of the art, and has been made and tested in accordance with the following directives and standards:

2014/35/EU

(formerly 2006/95/EC, valid until April 19, 2016)

Low Voltage Directive

Directive of the European Parliament and of the Council of February 26, 2014 on the harmonization of the laws of the Member States relating to the making available on the market of electrical equipment designed for use within certain voltage limits (recast).

2014/30/EU

(formerly 2004/108/EC, valid until April 19, 2016)

EMC Directive of the European Parliament and of the Council of February 26, 2014 on the harmonization of the laws of the Member States relating to electromagnetic compatibility (recast).

2006/42/EC

(Machinery Directive)

Directive of the European Parliament and of the Council of 17 May 2006 on machinery, and amending Directive 95/16/EC (recast).

Text Conventions

The following text conventions are used throughout the texts.

PATHS IN SOFTWARE

Paths are shown in software as “>”:

Example: Certus Control > Library > Valve.

REAL ACTIONS

If you are requested to take a real action, it is explicitly stated in the text. Listed actions are preceded by a triangle symbol.

Presentation example

► Right-click Layer and click the *Disable Layer* command.

ENGLISH TERMS

English terms used throughout the **CERTUS CONTROL** software use another font:

Example for Flush: `Flush` (Courier New, 12.0 points)

3.1.2 Safety

This section describes the essential safety requirements and provides information on CERTUS FLEX's safe operation.

Intended Use

The unit is intended for the following applications and uses:

- Fluid dispensing within dosage ranges starting from 50 nl for pharmacological, cytological, toxicological, chemical, biochemical, or genetical process and assay development
- ANSI-compliant well plate dispensing (low throughput dispensing)
- Use on an indoor laboratory bench
- Maximum temperature of fluids to be processed: 60°C

Any other or excessive use is deemed non-intended use. The manufacturer cannot be held liable for any damage that may be caused by it.

The intended use also includes the following:

- Compliance with all the instructions contained in this user manual.

Non-Intended Use

The following examples illustrate non-intended use (this list is not exhaustive):

- Installation, start-up, operation, and maintenance without prior instruction
- Operation outside the limits of technical specifications and beyond allowable environmental conditions
- Deactivation of safety devices
- Failure to comply with transportation, storage, installation, start-up, operation, maintenance, and repair instructions in the user manual
- Any alterations, modifications, or changes to the unit without the manufacturer's written approval
- Changes to the software without the manufacturer's written approval
- Improper repairs
- Use of incompatible materials
- Use of incompatible or damaged parts

Note

All warranty and liability claims are excluded if damage or injury is attributable to non-intended use.

Personnel Qualifications

All work on and involving the unit may only be performed by qualified and trained laboratory personnel who have read and understood this user manual.

The end user of the unit is responsible for:

- Selection, instruction, and training of personnel
- Continuous monitoring of work safety

REQUIREMENTS

- All work should only be performed by qualified laboratory personnel